

Yang-Jun Joo, Ph.D.

Quantitative Risk Assessment & Safety Data Scientist

Autonomous Vehicles | Statistical Modeling | Large-Scale Mobility Data

Orlando, FL (Open to relocation to Foster City, CA) | **Email** | Phone: +1 (407) 823-2841 | **LinkedIn** | **GitHub** | **Google Scholar**

Professional Summary: Quantitative safety researcher with 8+ years of experience developing statistical and machine-learning methods to measure autonomous-vehicle and transportation risk using large-scale driving, connected-vehicle, and traffic-signal data. Ph.D. in Transportation Engineering with experience leading funded safety research, building automated analytics pipelines, and analyzing human risk judgments in safety-critical driving scenarios.

TECHNICAL SKILLS

Programming & Data:	Python, SQL, R, pandas, NumPy, scikit-learn, PyTorch, Git, Linux, Apache Spark (PySpark), large-scale data pipelines.
Quantitative Methods:	Statistical modeling, probability, uncertainty quantification, conformal prediction, experimental design, regression analysis, spatiotemporal modeling, model validation.
Safety & Mobility:	Quantitative risk assessment, safety-performance metrics, autonomous-vehicle safety evaluation, surrogate safety measures (SSMs), ATSPM, connected-vehicle and telemetry data, human factors.

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar	University of Central Florida	Mar 2024 – Present
- Research autonomous driving safety evaluation and traffic safety analytics using advanced deep learning (including vision-language models) and conformal prediction. - Lead quantitative safety workstreams for FDOT-sponsored projects, coordinating with transportation agencies, faculty investigators, data engineers, and research staff to define metrics, validate analytical methods, and deliver operational safety insights. - Designed automated Python and SQL pipelines that transformed high-resolution ATSPM event logs and vehicle telemetry into reproducible safety-performance metrics and dashboard-ready hotspot indicators.		
Postdoctoral Researcher	Seoul National University	Mar 2023 – Feb 2024
- Conducted smart mobility research, leading traffic safety analytics and data-driven travel choice modeling. - Served as Principal Investigator (PI) on Sejong Science Fellowship (KRW 73.5 million), designing and analyzing human-subject studies comparing passenger risk judgments of autonomous and human-driven vehicles in near-miss and safety-critical scenarios. - Applied regression and heterogeneous-treatment analyses on dataset of 926 participants, 265 events, and 1,325 clips to identify how vehicle automation, event severity, and participant characteristics shaped perceived risk.		
Graduate Research Assistant	Seoul National University	Mar 2018 – Feb 2023
- Developed a Generalized Driving Risk Assessment for Highway Autonomous Vehicles Using a Risk-Field Approach for proactive collision avoidance. - Successfully secured competitive research funding (Global Ph.D. Fellowship - KRW 96.1 million) and published peer-reviewed research in Analytic Methods in Accident Research and Accident Analysis & Prevention.		

SELECTED RISK ASSESSMENT WORK

NPTOI: Network Level Proactive Traffic Operations	Co-Researcher Active Research Project
- Co-developed MMCAformer using connected-vehicle driving behavior data from four Florida freeways, reducing RMSE by 9.0%, MAE by 6.9%, and MAPE by 10.2%, while narrowing predictive intervals by 10.1–24.0%. - Co-developed a connected-vehicle traffic analytics framework covering 1,050 arterial links, estimating and predicting network-level delay and queue-length metrics under normal and abnormal traffic conditions.	

SELECTED PUBLICATIONS

- **Joo, Y.-J.**, Kim, E. J., Park, P. Y., & Kim, D. K. (2023). A Generalized Driving Risk Assessment Method for Autonomous Vehicles Using Field Theory. *Analytic Methods in Accident Research*, 37, 100251.
- **Joo, Y.-J.**, Kho, S. Y., Kim, D. K., & Park, H. C. (2022). A Data-Driven Bayesian Network for Probabilistic Crash Risk Assessment of Individual Driver with Traffic Violation and Crash Records. *Accident Analysis & Prevention*, 174, 106721.
- Han, L., Abdel-Aty, M., Kim, Y., **Joo, Y.-J.**, & Islam, Z. (2026). Macro-Micro Cross-Attention Transformer (MMCAformer) for Traffic Speed Prediction with Microscopic Connected Vehicle Data-based Driving Behavior. *IEEE Transactions on Intelligent Transportation Systems* (Under Revision).

EDUCATION

Seoul National University	Ph.D. in Civil & Environmental Engineering (Transportation Engineering)	Mar 2018 – Feb 2023
Seoul National University	B.S. in Civil & Environmental Engineering	Mar 2014 – Feb 2018